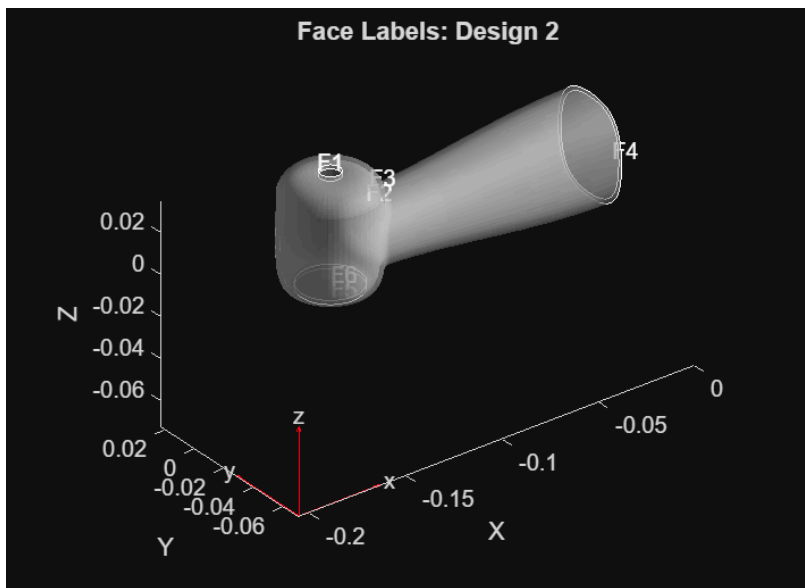
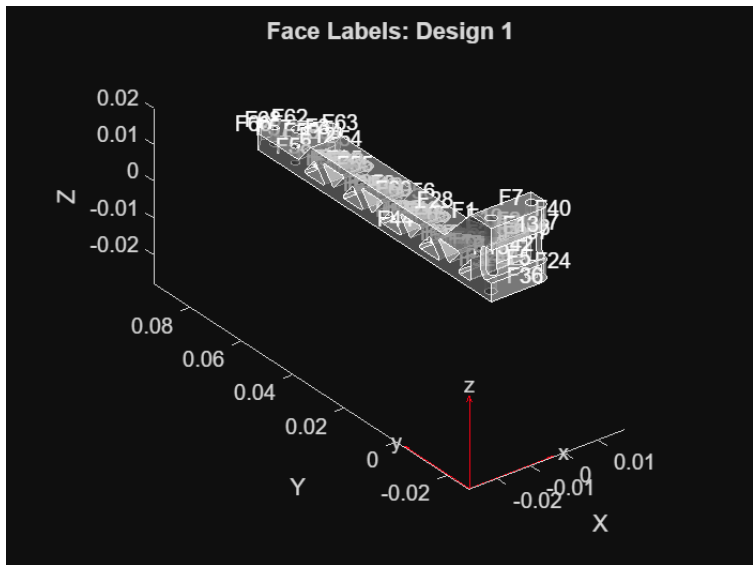
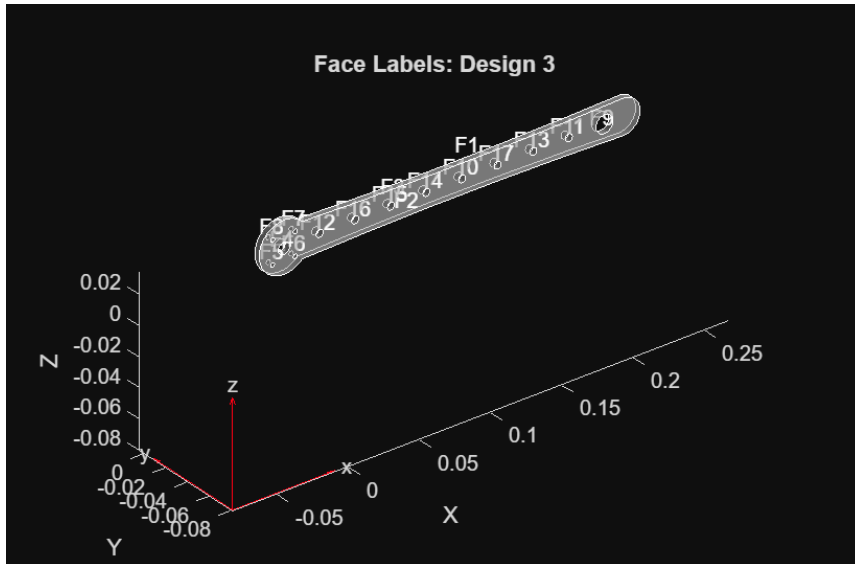


Design 1 (Jonah), Design 2 (Bao), Design 3 (Nathan)

Design	Material	DroneMass_kg	MaxPayload_kg	TWR	MeetsRequirement
{'Design 1'}	{["Carbon Fiber Composite (CFRP)"]}	1.048	0.952	2	true
{'Design 1'}	{["Aluminum Alloy"]}	1.084	0.916	2	true
{'Design 1'}	{["Fiberglass Composite (GFRP)"]}	1.06	0.94	2	true
{'Design 1'}	{["PLA Plastic"]}	1.038	0.962	2	true
{'Design 1'}	{["ABS Plastic"]}	1.032	0.968	2	true
{'Design 1'}	{["Wood (Birch)"]}	1.0184	0.9816	2	true
{'Design 2'}	{["Carbon Fiber Composite (CFRP)"]}	1.284	0.716	2	true
{'Design 2'}	{["Aluminum Alloy"]}	1.48	0.52	2	true
{'Design 2'}	{["Fiberglass Composite (GFRP)"]}	1.336	0.664	2	true
{'Design 2'}	{["PLA Plastic"]}	1.188	0.812	2	true
{'Design 2'}	{["ABS Plastic"]}	1.188	0.812	2	true
{'Design 2'}	{["Wood (Birch)"]}	1.108	0.892	2	true
{'Design 3'}	{["Carbon Fiber Composite (CFRP)"]}	1.144	0.856	2	true
{'Design 3'}	{["Aluminum Alloy"]}	1.24	0.76	2	true
{'Design 3'}	{["Fiberglass Composite (GFRP)"]}	1.168	0.832	2	true
{'Design 3'}	{["PLA Plastic"]}	1.112	0.888	2	true
{'Design 3'}	{["ABS Plastic"]}	1.092	0.908	2	true
{'Design 3'}	{["Wood (Birch)"]}	1.052	0.948	2	true





Motor thrust force = 9.810 N

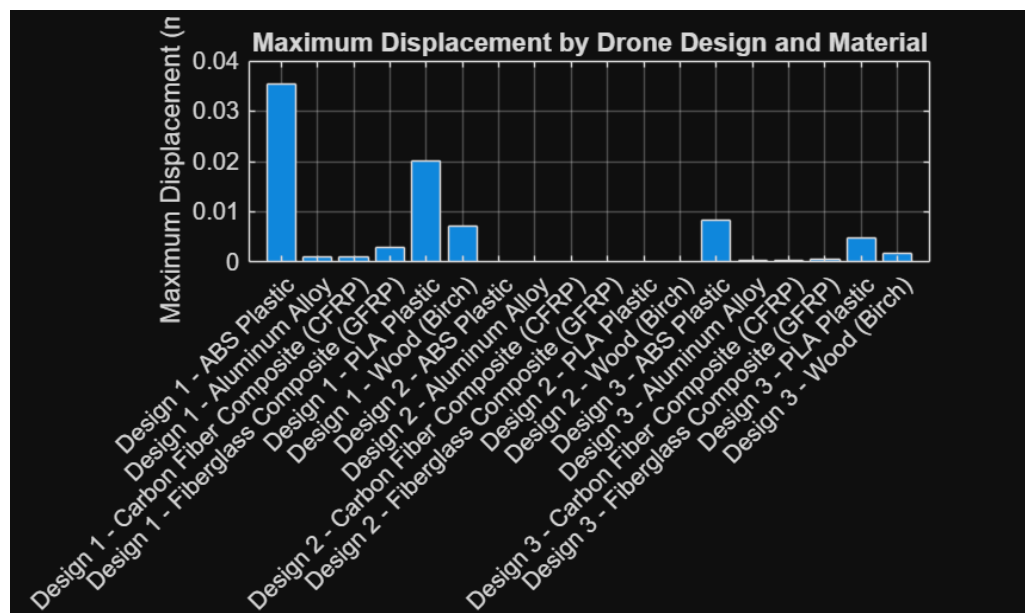
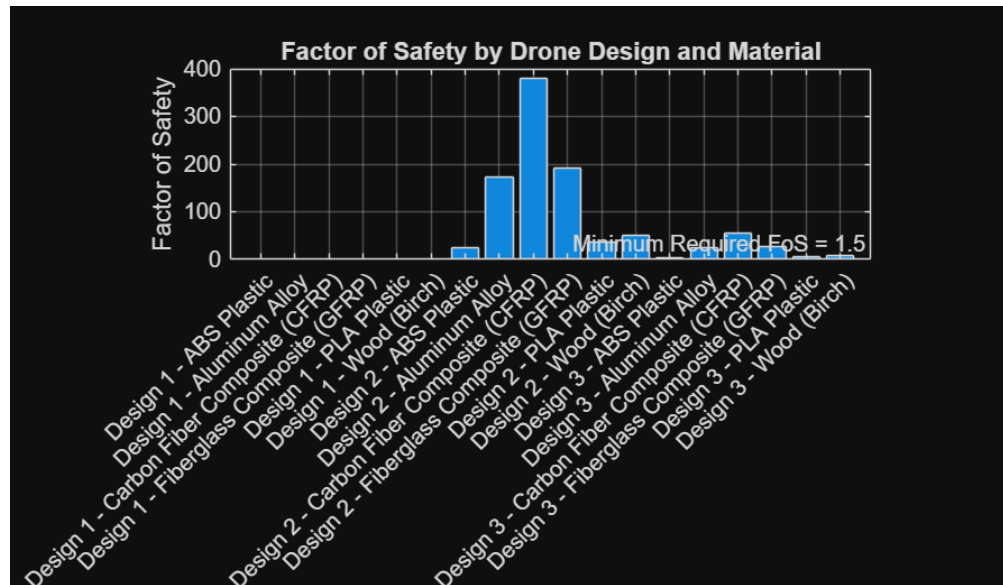
Motor weight force = 0.638 N

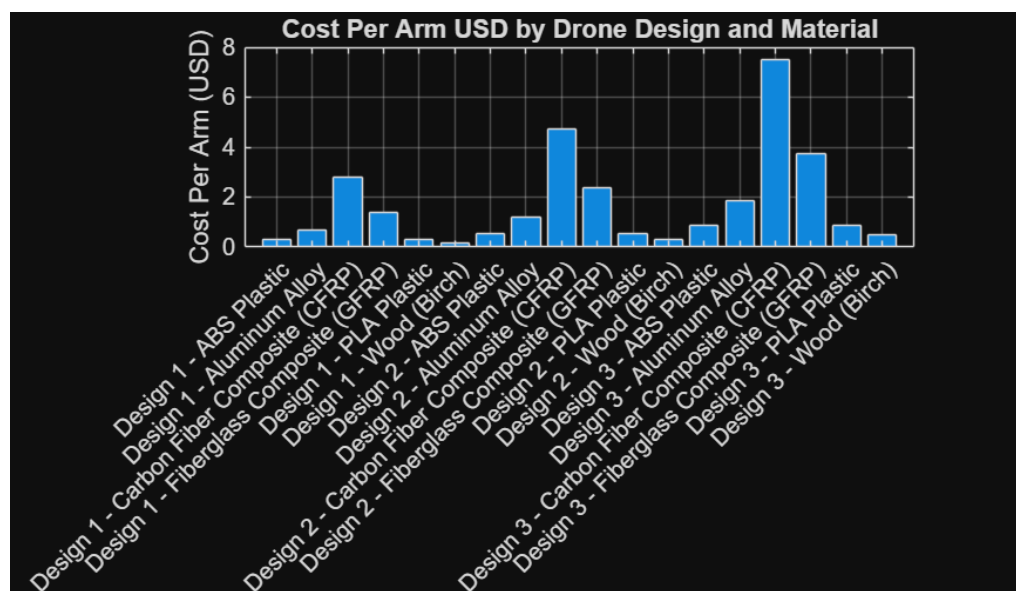
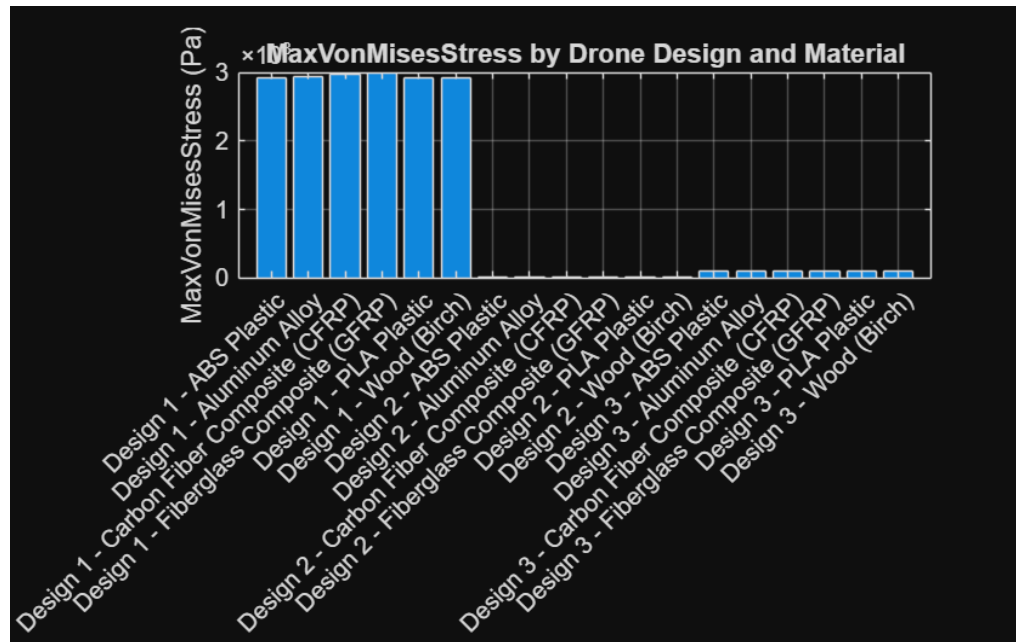
Net upward force = 9.172 N

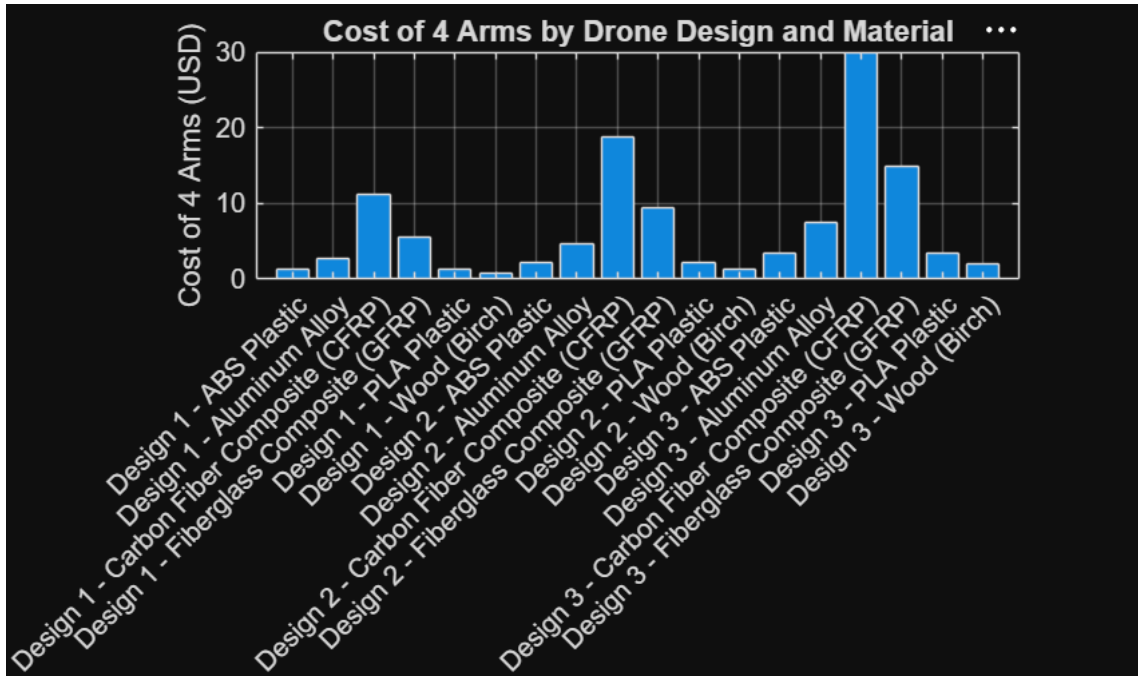
armLength_mm = 1x3

93.7400 157.0000 250.0000

Design	Material	MotorMountArea_m2	SurfaceTraction_N_m2	MaxDisplacement_m	MaxVonMisesStress_Pa	YieldStrength_Pa	FactorOfSafety	FEAStatus	CostPerArm
Design 1	"Carbon Fiber Composite (CFRP)"	1.2566e-05	7.2991e+05	0.0010105	2.9758e+08	6e+08	2.0163	"Good Design Range"	2.8122
Design 1	"Aluminum Alloy"	1.2566e-05	7.2991e+05	0.001026	2.9455e+08	2.75e+08	0.93364	"Fail"	0.70305
Design 1	"Fiberglass Composite (GFRP)"	1.2566e-05	7.2991e+05	0.0028268	2.9968e+08	3e+08	1.0011	"Fail"	1.4061
Design 1	"PLA Plastic"	1.2566e-05	7.2991e+05	0.02023	2.9167e+08	6e+07	0.20571	"Fail"	0.32809
Design 1	"ABS Plastic"	1.2566e-05	7.2991e+05	0.035403	2.9261e+08	4e+07	0.1367	"Fail"	0.32809
Design 1	"Wood (Birch)"	1.2566e-05	7.2991e+05	0.0070806	2.9261e+08	8e+07	0.2734	"Fail"	0.18748
Design 2	"Carbon Fiber Composite (CFRP)"	7.854e-05	1.1679e+05	2.593e-06	1.5796e+06	6e+08	379.83	"Overbuilt"	4.71
Design 2	"Aluminum Alloy"	7.854e-05	1.1679e+05	2.6328e-06	1.5937e+06	2.75e+08	172.55	"Overbuilt"	1.1775
Design 2	"Fiberglass Composite (GFRP)"	7.854e-05	1.1679e+05	7.2537e-06	1.5699e+06	3e+08	191.1	"Overbuilt"	2.355
Design 2	"PLA Plastic"	7.854e-05	1.1679e+05	5.1909e-05	1.6069e+06	6e+07	37.339	"Overbuilt"	0.5495
Design 2	"ABS Plastic"	7.854e-05	1.1679e+05	9.0846e-05	1.6026e+06	4e+07	24.959	"Overbuilt"	0.5495
Design 2	"Wood (Birch)"	7.854e-05	1.1679e+05	1.8169e-05	1.6026e+06	8e+07	49.918	"Overbuilt"	0.314
Design 3	"Carbon Fiber Composite (CFRP)"	0.0001131	81101	0.00023768	1.0813e+07	6e+08	55.488	"Overbuilt"	7.5
Design 3	"Aluminum Alloy"	0.0001131	81101	0.00024172	1.0726e+07	2.75e+08	25.638	"Overbuilt"	1.875
Design 3	"Fiberglass Composite (GFRP)"	0.0001131	81101	0.00066434	1.0861e+07	3e+08	27.621	"Overbuilt"	3.75
Design 3	"PLA Plastic"	0.0001131	81101	0.0047758	1.0616e+07	6e+07	5.6521	"Overbuilt"	0.875
Design 3	"ABS Plastic"	0.0001131	81101	0.0083519	1.0656e+07	4e+07	3.7538	"Good Design Range"	0.875
Design 3	"Wood (Birch)"	0.0001131	81101	0.0016704	1.0656e+07	8e+07	7.5077	"Overbuilt"	0.5







Combined Payload, TWR, FEA, Stress, and Cost Results:									
Design	Material	MotorMountArea_m2	SurfaceTraction_N_m2	MaxDisplacement_m	MaxVonMisesStress_Pa	YieldStrength_Pa	FactorOfSafety	FEAStatus	CostPerArm
"Design 1"	"Carbon Fiber Composite (CFRP)"	1.2566e-05	7.2991e+05	0.0010105	2.9758e+08	6e+08	2.0163	"Good Design Range"	2.8122
"Design 1"	"Aluminum Alloy"	1.2566e-05	7.2991e+05	0.001026	2.9455e+08	2.75e+08	0.93364	"Fail"	0.70305
"Design 1"	"Fiberglass Composite (GFRP)"	1.2566e-05	7.2991e+05	0.0028268	2.9968e+08	3e+08	1.0011	"Fail"	1.4061
"Design 1"	"PLA Plastic"	1.2566e-05	7.2991e+05	0.02023	2.9167e+08	6e+07	0.20571	"Fail"	0.32809
"Design 1"	"ABS Plastic"	1.2566e-05	7.2991e+05	0.035403	2.9261e+08	4e+07	0.1367	"Fail"	0.32809
"Design 1"	"Wood (Birch)"	1.2566e-05	7.2991e+05	0.0070806	2.9261e+08	8e+07	0.2734	"Fail"	0.18748
"Design 2"	"Carbon Fiber Composite (CFRP)"	7.854e-05	1.1679e+05	2.593e-06	1.5796e+06	6e+08	379.83	"Overbuilt"	4.71
"Design 2"	"Aluminum Alloy"	7.854e-05	1.1679e+05	2.6328e-06	1.5937e+06	2.75e+08	172.55	"Overbuilt"	1.1775
"Design 2"	"Fiberglass Composite (GFRP)"	7.854e-05	1.1679e+05	7.2537e-06	1.5699e+06	3e+08	191.1	"Overbuilt"	2.355
"Design 2"	"PLA Plastic"	7.854e-05	1.1679e+05	5.1909e-05	1.6069e+06	6e+07	37.339	"Overbuilt"	0.5495
"Design 2"	"ABS Plastic"	7.854e-05	1.1679e+05	9.0846e-05	1.6026e+06	4e+07	24.959	"Overbuilt"	0.5495
"Design 2"	"Wood (Birch)"	7.854e-05	1.1679e+05	1.8169e-05	1.6026e+06	8e+07	49.918	"Overbuilt"	0.314
"Design 3"	"Carbon Fiber Composite (CFRP)"	0.0001131	81101	0.00023768	1.0813e+07	6e+08	55.488	"Overbuilt"	7.5
"Design 3"	"Aluminum Alloy"	0.0001131	81101	0.00024172	1.0726e+07	2.75e+08	25.638	"Overbuilt"	1.875
"Design 3"	"Fiberglass Composite (GFRP)"	0.0001131	81101	0.00066434	1.0861e+07	3e+08	27.621	"Overbuilt"	3.75
"Design 3"	"PLA Plastic"	0.0001131	81101	0.0047758	1.0616e+07	6e+07	5.6521	"Overbuilt"	0.875
"Design 3"	"ABS Plastic"	0.0001131	81101	0.0083519	1.0656e+07	4e+07	3.7538	"Good Design Range"	0.875
"Design 3"	"Wood (Birch)"	0.0001131	81101	0.0016704	1.0656e+07	8e+07	7.5077	"Overbuilt"	0.5

Best overall design-material combination:									
Design	Material	MotorMountArea_m2	SurfaceTraction_N_m2	MaxDisplacement_m	MaxVonMisesStress_Pa	YieldStrength_Pa	FactorOfSafety	FEAStatus	CostPerArm
"Design 1"	"Carbon Fiber Composite (CFRP)"	1.2566e-05	7.2991e+05	0.0010105	2.9758e+08	6e+08	2.0163	"Good Design Range"	2.8122

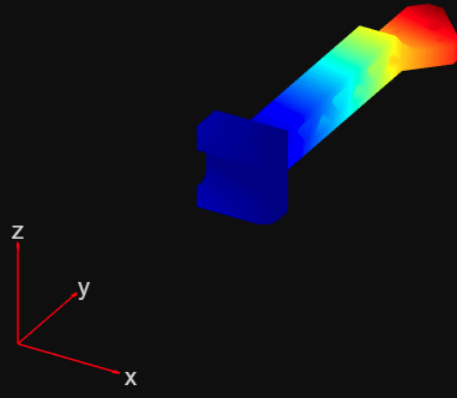
Ranked overall results:									
Design	Material	MotorMountArea_m2	SurfaceTraction_N_m2	MaxDisplacement_m	MaxVonMisesStress_Pa	YieldStrength_Pa	FactorOfSafety	FEAStatus	CostPerArm
"Design 1"	"Carbon Fiber Composite (CFRP)"	1.2566e-05	7.2991e+05	0.0010105	2.9758e+08	6e+08	2.0163	"Good Design Range"	2.8122
"Design 3"	"Wood (Birch)"	0.0001131	81101	0.0016704	1.0656e+07	8e+07	7.5077	"Overbuilt"	0.5
"Design 3"	"ABS Plastic"	0.0001131	81101	0.0083519	1.0656e+07	4e+07	3.7538	"Good Design Range"	0.875
"Design 2"	"Wood (Birch)"	7.854e-05	1.1679e+05	1.8169e-05	1.6026e+06	8e+07	49.918	"Overbuilt"	0.314
"Design 3"	"PLA Plastic"	0.0001131	81101	0.0047758	1.0616e+07	6e+07	5.6521	"Overbuilt"	0.875
"Design 3"	"Carbon Fiber Composite (CFRP)"	0.0001131	81101	0.00023768	1.0813e+07	6e+08	55.488	"Overbuilt"	7.5
"Design 3"	"Fiberglass Composite (GFRP)"	0.0001131	81101	0.00066434	1.0861e+07	3e+08	27.621	"Overbuilt"	3.75
"Design 2"	"ABS Plastic"	7.854e-05	1.1679e+05	9.0846e-05	1.6026e+06	4e+07	24.959	"Overbuilt"	0.5495
"Design 2"	"PLA Plastic"	7.854e-05	1.1679e+05	5.1909e-05	1.6069e+06	6e+07	37.339	"Overbuilt"	0.5495
"Design 3"	"Aluminum Alloy"	0.0001131	81101	0.00024172	1.0726e+07	2.75e+08	25.638	"Overbuilt"	1.875
"Design 2"	"Carbon Fiber Composite (CFRP)"	7.854e-05	1.1679e+05	2.593e-06	1.5796e+06	6e+08	379.83	"Overbuilt"	4.71
"Design 2"	"Fiberglass Composite (GFRP)"	7.854e-05	1.1679e+05	7.2537e-06	1.5699e+06	3e+08	191.1	"Overbuilt"	2.355
"Design 2"	"Aluminum Alloy"	7.854e-05	1.1679e+05	2.6328e-06	1.5937e+06	2.75e+08	172.55	"Overbuilt"	1.1775

Best Design Overall was Design 1 (Jonah), with Carbon Fiber Composite.

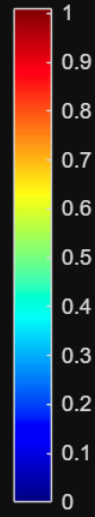
Figures below visualized displacement magnitude and von Mises stress on the design.

DesignPlot = 1

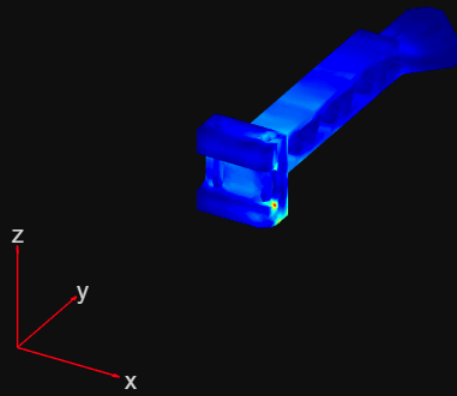
Displacement Magnitude: Design 1



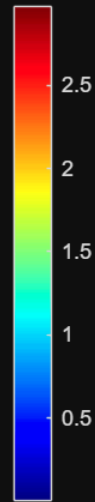
$\times 10^{-3}$



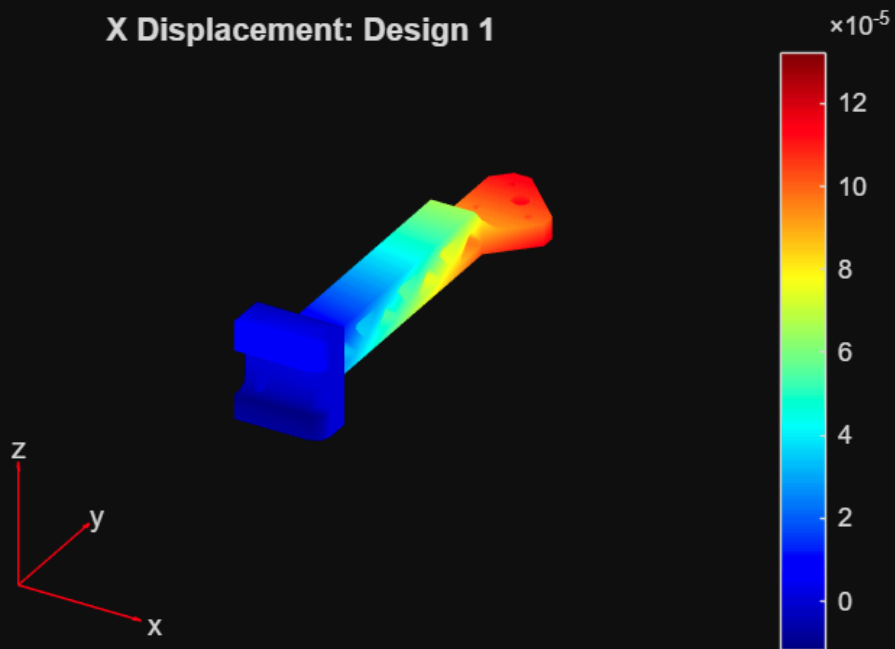
Von Mises Stress: Design 1



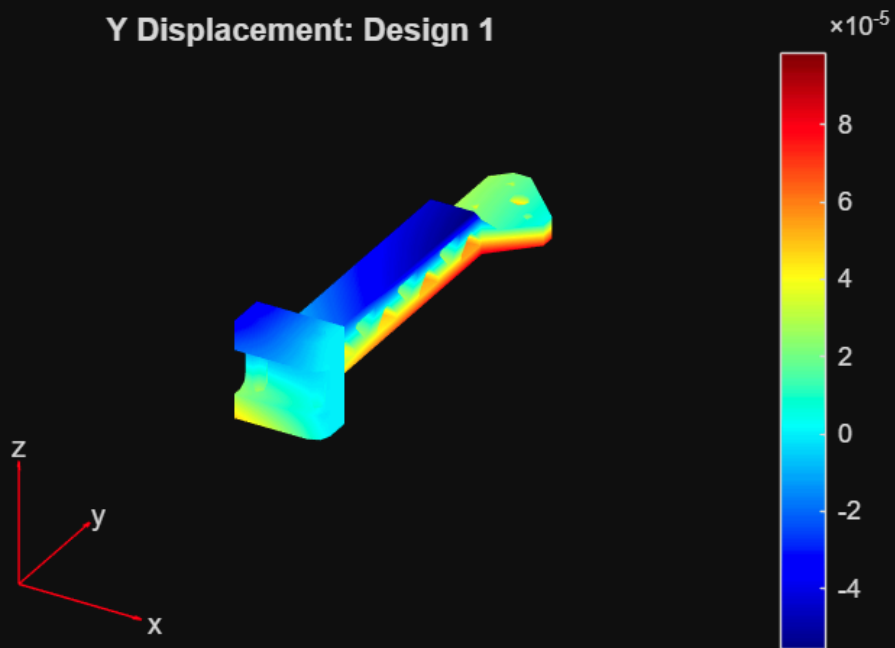
$\times 10^8$



X Displacement: Design 1



Y Displacement: Design 1



Z Displacement: Design 1

